

Ritabrata Chakraborty

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EDUCATION

University of Pennsylvania

M.S.E. in Robotics, GPA: -/4

Pennsylvania, USA

Aug '26 – May '28

Birla Institute of Technology and Science, Pilani (BITS Pilani)

B.E. in Mechanical Engineering — Minor in Data Science, CGPA: 8.57/10

Rajasthan, India

Oct '22 – May '26

- **Institute Merit-cum-Need Scholarship** — Consecutive Recipient (Top 8% of Students)

PUBLICATIONS & PEER REVIEWS

Journal Publications

- R. Chakraborty and K. Kishore, “*Navigating Without Satellites: A Critical Survey of GNSS-Denied UAV Navigation.*” Submitted to **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026. [🔗](#)
- R. Chakraborty and K. Kishore, “*Multi-Critic Off-Policy Reinforcement Learning for Quadrotor Velocity Control in Unknown Indoor Environments.*” Manuscript in preparation for **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026.

Conference Publications

- Y. Wang, H. He, Y. Cao, J. Liang, R. Chakraborty, and G. Sartoretti, “*CogniPlan: Uncertainty-Guided Path Planning with Conditional Generative Layout Prediction.*” Accepted at **Conference on Robot Learning (CoRL)**, 2025; published in **Proceedings of Machine Learning Research (PMLR)**. [🔗](#)
- R. Chakraborty, T. Mian, and P. Kundu, “*An Efficient Approach for Synthetic Data Generation and Fault Diagnosis for Rotating Machinery.*” Accepted at **Prognostics and System Health Management Conference (PHM)**, 2025; published in **IET Conference Proceedings**. [🔗](#)
- K. Kishore and R. Chakraborty, “*Path Planning of Low-Altitude UAV for Tree Canopy Tracking and Orchard Monitoring.*” Publication awaiting intellectual property clearance, 2025. [🔗](#)

Poster Presentations

- R. Chakraborty, “*High-Velocity Impact Testing and Analysis of Layered Metal-Composite Shields Using Electromagnetic Railgun.*” Presented as poster at **Pulsed Power Science, Technology and Application (PPSTA)**, Visakhapatnam, India, Sep 2024.

Peer Reviews

- “*GS-SDE: 3D Gaussian Splatting-based Adaptive Scene Modeling in Semi-Dynamic Environments.*” Peer review completed for **IEEE Transactions on Automation Science and Engineering (T-ASE)**, 2026.
- “*Limits of Wearable Gait Analysis for Age Classification in Young Adults: Anthropometry Over Age.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- “*Generative AI and Computer Vision Based Multi-Assessment System for Early Prediction and Diagnosis of Autism Spectrum Disorder (ASD).*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- “*AquaNode: Design and Implementation of a Low-Cost Automated Water Tank Monitoring and Valve Control System Using Arduino and Ultrasonic Sensing.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- “*A Research Paper on IoT-Based Robot for Agriculture.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- Selected as a peer reviewer for **IEEE Gujarat Section Conference**, 2026.

EXPERIENCE – ROBOTICS & COMPUTER VISION

Vision-Language Grounded Multi-Robot Coordination and Navigation

Supervisor: **Dr. Avinash Gautam**, Associate Professor, BITS Pilani

May '25 – Aug '25

- Surveyed literature on vision-language grounded multi-robot coordination and VLMaps-based spatial representations.
- Reviewed decentralized navigation strategies fusing 3D visual-language maps with LLM-driven instruction parsing.

Multi-Critic Decomposition for Autonomous Indoor Navigation [🔗](#)

Supervisor: **Dr. Kaushal Kishore**, Principal Scientist, CSIR-CEERI

Jan '26 – May '26

- Integrated three specialized critics for forward progress, lateral avoidance, and angular alignment with distinct reward signals.

- Deployed multi-critic TD3 with prioritized replay achieving 94.2% success rate and 22% lower collision rate versus baseline.

Vision-Attention-Driven Autonomous Navigation with Semantic Understanding

Supervisor: [Dr. Guillaume Sartoretti](#), Assistant Professor, MARMoT Lab, NUS

Aug '25 – Dec '25

- Enhanced [CogniPlan](#) with cross-attention between frontier and node embeddings to enrich node representations.
- Incorporated [Visual Navigation Transformer \(ViNT\)](#) to capture semantic context for adaptive exploration strategies.

Design and Development of Smart Automated Field Hockey Ball Launcher [↗](#)

Supervisor: [Dr. Mani Shankar Dasgupta](#), Associate Professor, BITS Pilani

Apr '24 – Oct '25

Physics-Based Launcher Mechanical Design and FEM Analysis

- Fabricated programmable launcher achieving 150 km/h using CFRP composite and stainless steel components.
- Performed FEM analysis and CAD using Fusion 360, achieving safety factor 2.07 at 496 RPM.

Computer Vision Pipeline for 3D Trajectory Reconstruction and Shot Classification

- Devised 3D ball localization from monocular videos using diameter-based depth and [PnLCalib](#) (0.76m RMSE).
- Constructed TCN-attention classifier fusing trajectory and statistical features, achieving 96.4% classification accuracy.

Uncertainty-Guided Path Planning via Conditional Layout Prediction [↗](#)

Supervisor: [Dr. Guillaume Sartoretti](#), Assistant Professor, MARMoT Lab, NUS

Mar '25 – Aug '25

- Adapted multiple image-based exploration planners into ROS pipelines for evaluation in the [CMU Exploration Environment](#).
- Supported the development and benchmarking of [CogniPlan](#), achieving up to 17.7% shorter exploration paths and 3.9% improved navigation efficiency over state-of-the-art baselines across 100+ unseen maps.

Monocular Vision-Based UAV Navigation for Orchard Monitoring [↗](#)

Supervisor: [Dr. Kaushal Kishore](#), Principal Scientist, CSIR-CEERI

Jan '24 – Feb '25

- Engineered a UAV-based orchard monitoring system using YOLOv11 (Box mAP50: 95.5%, Mask mAP50: 96.5%).
- Implemented B-spline trajectory logic and coded custom yaw-roll controller to minimize drift under wind.

Image-Based Visual Servoing for Automated Radar Control and UAV Tracking

Supervisor: [Dr. Rishi Verma](#), Head (ESDS), PPEMD, BARC

Jun '24 – Jul '24

- Established automated radar alignment using YOLOv8 and DeepSORT, improving tracking recall to 91% and speed by 13%.
- Interfaced a PLC-based actuation system enabling continuous UAV lock-in within $\pm 5^\circ$ angular error.

EXPERIENCE – GENERATIVE AI & PREDICTIVE MAINTENANCE

Synthetic Data Generation for Bearing Fault Diagnosis Under Data Scarcity

Supervisor: [Dr. Pradeep Kundu](#), Assistant Professor, KU Leuven

Sep '24 – May '26

Auxiliary Classifier WGAN-GP for Time-Series Sensor Data Generation [↗](#)

- Built ACWGAN-GP with TCN discriminator for minority fault class augmentation reaching $\sim 100\%$ classification accuracy.
- Assessed synthetic data using PCC, Cosine Similarity, MMD, and KL Divergence on FFT spectra.

Physics-Informed Generative Modeling for Bearing Fault Diagnosis Under Data Scarcity

- Formulated physics-informed cVAE-GAN and cVQ-GAN frameworks isolating latent-space effects in CWT scalogram synthesis.
- Validated [LiteFormer](#) classification gains across Density, Coverage, and CMMD metrics.

EXPERIENCE – MECHANICAL SIMULATION & BEHAVIORAL RESEARCH

Finite Element Analysis of Running Track Surfaces

Supervisor: [Dr. Mani Shankar Dasgupta](#), Associate Professor, BITS Pilani

Apr '24 – Sep '25

- Modeled surface materials using hyperelastic formulations, evaluating restitution across 3 sports disciplines.
- Proposed optimized track designs improving durability by 23% based on ANSYS fatigue analysis.

Primal World Beliefs in Engineering Students [↗](#)

Supervisor: [Dr. Rajneesh Choubisa](#), BITS Pilani

Aug '24 – Dec '24

- Conducted a primal world beliefs study surveying engineering undergraduates on core worldview dimensions.
- Replicated established population-level findings, validating trends against the original general-population study.

Impact Testing and Analysis of Metal-Composite Shields Using Electromagnetic Railguns

Supervisor: [Dr. Rishi Verma](#), Head (ESDS), PPEMD, BARC

Jun '24 – Jul '24

- Characterized 3 material configurations via the Johnson-Cook model for high-strain-rate impact response.
- Verified simulation results through railgun tests at 3.2 MJ energy, achieving $< 7\%$ deviation from FEM predictions.

INDUSTRY EXPERIENCE

ML Intern Uber (AI Solutions)	
Supervisor: <i>Mr. Siddarth Malreddy, Tech Lead Manager & Mr. Ishan Nigam, Senior ML Engineer, Uber</i>	<i>Jul '25 – Dec '25</i>
- Streamlined object tracking in uLabel for RGB/IR videos, flagging object annotation anomalies via XGBoost (mAP50: 75%). - Screened object-association errors with a TrackTrack-based cross-frame consistency check (F1: 88%).	
Subject Matter Expert Knowledge Cultivator	
<i>Self-supervised</i>	<i>Nov '23 – Feb '24</i>
- Resolved numerical problems for student clients using MATLAB, CAD, and Python. - Authored academic content across diverse research topics.	

PROJECTS

Vision-Guided Robotic Arm for Simulated Oral Cancer Screening	<i>Aug '25 – Dec '25</i>
- Simulated a KUKA KR 150 R3100 arm in MATLAB/Simulink for end-effector guidance inside a modeled oral cavity. - Trained a vision model to detect cancerous regions, localizing the end effector through ORB-SLAM3 tracking.	
Autonomous Drone Navigation MathWorks Global Student Drone Challenge 2025	<i>Mar '25 – Apr '25</i>
- Designed vision-based control for Parrot Mambo drone using masking, ray-tracing, and closed-loop yaw control. - Optimized speed control and zone-based auto-landing to reduce track completion time.	
Air Quality Index Forecasting Using Time Series Models ↗	<i>Nov '24 – Nov '24</i>
- Benchmarked AR, MA, ARMA, ARIMA, and SARIMA models forecasting daily and weekly AQI for Jaipur and Kolkata. - Diagnosed stationarity with ADF and KPSS tests, tuning lag orders via ACF/PACF and scoring on R², MAE, and RMSE.	
3D Lunar Surface Modeling from DEM Data ISRO Immersion Challenge 2024 ↗	<i>Jul '24 – Jul '24</i>
- Created lunar terrain models from TMC DEM data with curvature extraction, rendered live in UE5. - Automated data import and heightmap transformation via a custom plugin, processing each tile in under 10s.	
Cricket Team Optimization AMEX Campus Challenge 2024 (Product Track) ↗	<i>Jun '24 – Jul '24</i>
- Processed and cleaned 100K+ datapoints to identify anomalies, outliers, and consistency trends. - Generated optimal lineups using statistical heuristics, achieving a 14% gain over baseline strategies.	
ExoMy Rover Navigation and UR3 Arm Motion Planning ERC 2023 Remote ↗	<i>Apr '23 – Sep '23</i>
- Navigated ExoMy rover using ArUco detection, Ackermann steering, and spot turns for autonomous hazard avoidance. - Calibrated UR3 arm with MoveIt and OMPL planner for collision-free manipulation (98% success).	

AWARDS & ACHIEVEMENTS

Finalist — Meta PyTorch OpenEnv Hackathon (India) ↗	<i>Apr '26</i>
3rd Place — MathWorks Global Drone Student Challenge 2025	<i>Mar '25</i>
Finalist — ISRO Immersion Challenge (AI for Space & Geospatial Innovation) ↗	<i>Jul '24</i>
Top 15 Overall, Top 5 in College — American Express Campus Challenge 2024 ↗	<i>Jul '24</i>
5th Place & Best Maintenance Award — European Rover Challenge 2023 (Remote Finals)	<i>Sep '23</i>

TEACHING

Teaching Assistant	
<i>CIS 4190/5190: Applied Machine Learning, University of Pennsylvania</i>	<i>Aug '26 – Dec '26</i>
<i>CS F320: Foundations of Data Science, BITS Pilani</i>	<i>Jan '26 – May '26</i>
<i>ME F218: Advanced Mechanics of Solids, BITS Pilani</i>	<i>Jan '25 – May '25</i>
<i>ME F216: Materials Science and Engineering, BITS Pilani</i>	<i>Sep '24 – Dec '24</i>
- Assisted 100+ students in labs and tutorials, clarifying concepts and linking theory to practical applications. - Analyzed assignments and facilitated faculty in delivering high-impact teaching sessions.	

LEADERSHIP & VOLUNTEERING

Deputy Director of Academic Programming <i>Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania</i>	<i>Aug '26 – Present</i>
Vice-President <i>South Asian Association (Rangoli), University of Pennsylvania</i>	<i>Aug '26 – Present</i>
President & Secretary <i>Mechanical Engineering Association (MEA), BITS Pilani</i> - Coordinated 10+ events and career sessions for 300+ students, facilitating technical exposure and alumni interaction. - Managed production of 500+ merchandise items and led outreach, boosting student participation by 20%.	<i>Jun '24 – May '26</i>
President & Tech Fest Coordinator <i>Indian Society of Heating, Refrigerating, and Air Conditioning Engineers (ISHRAE), BITS Pilani</i> - Led a team of 25+ members to organize 4+ technical workshops with HVAC industry experts. - Hosted 3 competitions and networking events, engaging 200+ students in HVAC innovation and awareness.	<i>Oct '24 – Jul '25</i>
Project Manager <i>Tinkerer's Lab (TL), BITS Pilani</i> - Supervised 5 interdisciplinary robotics teams (30+ members) on Micromouse and Hexapod projects. - Oversaw lab resources, conducted weekly reviews, and mentored 50+ students in hands-on technical skills.	<i>May '24 – Jul '25</i>
Treasurer <i>Bengali Cultural Association (Moruchhaya), BITS Pilani</i> Allocated funds and maintained financial records for merchandise sales, cultural programs, and annual events.	<i>May '24 – Jul '25</i>
Volunteer <i>Junoon, National Service Scheme (NSS), BITS Pilani</i> Organized sports and recreational competitions for specially abled children from 25+ colleges.	<i>Sep '23 – Sep '23</i>

PROFESSIONAL ACTIVITIES

Graduate Student Member — IEEE	<i>2026</i>
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COMPETITIVE EXAMINATIONS

GRE: 317 (Quantitative: 168, Verbal: 149, AWA: 3)	<i>2025</i>
TOEFL iBT: 108 (Reading: 27, Listening: 30, Speaking: 23, Writing: 28)	<i>2025</i>
JEE Advanced: All India Rank 14372	<i>2022</i>

TECHNICAL SKILLS & COURSEWORK

Relevant Coursework: Machine Learning, Deep Learning, Foundations of Data Science, Applied Statistical Methods, Linear Algebra, Computer Programming, Engineering Optimization, Differential Equations, Design of Machine Elements, Digital Twins
Programming Languages: Python, C++, C, C#, Java, Bash (Shell)
Robotics & Simulation: ROS (with Gazebo, Rviz), MAVROS, Navigation Stack, MoveIt, AirSim, Unity, Unreal Engine, MATLAB, Simulink, QGIS
AI & Data Science: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, OpenCV, Open3D, NumPy, Pandas, SciPy, StatsModels, Matplotlib, Seaborn, Weights & Biases (W&B), Excel
Cloud & MLOps: AWS (S3, EC2, SageMaker), Azure (Foundry), Google Cloud Platform (GCP), Docker, Git, GitHub Actions
Hardware & Embedded Systems: NVIDIA Jetson (Nano, Orin), Raspberry Pi, Arduino, IMUs, Stereo Camera, 3D LiDAR
CAD & Mechanical Simulation: ANSYS Mechanical, SolidWorks, Fusion 360
Leadership & Communication: Event Planning, Team Management, Budget Allocation, Mentorship, Cross-Functional Coordination, Public Speaking, Stakeholder Outreach, Technical Writing
Productivity Software: Microsoft Office (Excel, PowerPoint, Word), Google Workspace